# ProbOS — Month 3, Week 12
# Automotive / EV Safety Positioning

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## Week 12 goal

Produce accurate, well-researched positioning materials mapping `BatteryModel2Cell`'s already-validated capabilities to ISO 26262's automotive functional-safety framework.

**Critical distinction from Weeks 9–11, stated up front:** this is NOT an engineering week. No new model, no new code, and no new validation dataset are required – the underlying battery model is already Validated (Kim 2007, Sobol $S_1 = 0.457$ for Ea_SEI). The work here is entirely research and positioning: correctly understanding ISO 26262's actual terminology and requirements (ASIL levels, Hazard Analysis and Risk Assessment / HARA, safety case documentation) well enough to map our existing capabilities onto them accurately, without overclaiming compliance or certification we have not actually obtained. Exit criteria are "materials exist and are accurate," not "tests pass."

## Day-by-day plan

### Day 1 – ISO 26262 research (before writing any positioning material)

- Research ISO 26262's actual structure: ASIL classification, the Hazard Analysis and Risk Assessment (HARA) process, and what documentation a safety case actually requires – from real, citable sources, not assumed from general familiarity
- Identify precisely which parts of `BatteryModel2Cell`'s existing output (Sobol sensitivity indices, provenance-traced tail-risk outcomes, Monte Carlo percentile trajectories) map genuinely onto HARA-style hazard identification – and be explicit about anything that does NOT map cleanly
- Confirm: are we claiming ProbOS *could support* an ISO 26262 safety case, or that it *has been used in* one? Only the former is currently true and must be the only claim made

### Day 2 – Draft the automotive/EV positioning brief

- A dedicated technical brief for an automotive/EV safety engineering audience, explaining what `BatteryModel2Cell` already demonstrates and how it maps to ISO 26262 concepts
- Explicitly reuse the REAL, already-computed results (Kim 2007 validation, $S_1 = 0.457$ for Ea_SEI, the P95-worst-case-cell result) – no new numbers are invented for this audience

### Day 3 – Outreach materials

- Draft an outreach email template for automotive/EV safety engineers, analogous to `docs/enterprise_pilot_ema` but addressed to this specific audience
- Keep the ask concrete and modest (a conversation to confirm or correct the assumed pain point), not an overreaching pitch

### Day 4 – Cross-check against the honest open questions

- Re-read `docs/vision/main.tex`'s stated open questions and confirm the Week 12 materials do not implicitly resolve or paper over either of them

### Day 5 – Review pass

- Full read-through checking for any claim not directly traceable to either an actual computed result already in the codebase, or an accurately cited external source

### Day 6 – Confirm no regressions

- Run `pre_week_audit.sh` regardless, since this is a docs-only week – confirms nothing was accidentally broken

### Day 7 – Documentation + retrospective

- Update `docs/vision/main.tex`'s automotive section to note positioning materials are complete
- Retrospective appended to this document once complete

## Exit criteria

Accurate, well-researched automotive/EV positioning materials exist, citing only real ISO 26262 concepts and real, already-computed ProbOS results – with no overclaimed certification, compliance, or practitioner validation that has not actually occurred.